WEEK 5 · DAY 2 · RULE BOOK

Forecasting Baselines

Follow it step by step. Every common error and its fix is here.

Your goal today: lock the last 8 weeks of NOC tickets away, forecast them four cheap ways, score every forecast with the same ruler (MAE), then build the two-line forecast that beats them all. **14 green ✅ ticks** to collect. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open `W5D2-exercise` in Colab → **File** → **Save a copy in Drive**.
- ☐ Run **STEP 0**. Wait for ✅ `Ready!`
- ☐ Run every cell in order. Each ✍️ cell depends on the cells above it.
- ☐ The ▶️ cell with `def mae(...)` (top of Part B) must run before the scoreboards.

The four baselines (and the fifth that wins)

Baseline	Grandma says	One line of the function	Forecast for all 8 test weeks
🐵 **Naive**	"Same as last week"	`series.iloc[-1]`	74 every week
📅 **Seasonal naive**	"Same week last year"	`series.iloc[-52:]`	42, 42, 44, 42, 53, 51, 52, 54
🧱 **Moving average**	"The average of the last 8 weeks"	`series.iloc[-8:].mean()`	77 every week
🍚 **Exponential smoothing**	"A running level"	`alpha * v + (1 - alpha) * level`	75.6 every week (α = 0.3)
🎯 **Rate seasonal naive**	"Last year's rate × today's network"	`rate = series / terminals`	56.4, 56.1, 58.5, 55.6, 69.8, 66.8, 67.7, 70.0

MAE (mean absolute error) = the average size of the misses, in tickets. *"On a typical week we were off by N tickets."* Lower is better.

The exercise, step by step

Task	What you do	The tool you need
A1	Load the weeks with the right frequency	`FREQ = "W-MON"` (in quotes)
A2	Lock the last weeks away	`H = 8`
B1	Naive: the last value	`series.iloc[-1]`

Task	What you do	The tool you need
B2	Seasonal naive: the season length	`period=52`
B3	Moving average: the window	`window=8`
B4	Exponential smoothing: what stays of the old level	`(1 - alpha)`
B5	The scoreboard: compare with what really happened	`COMPARE_WITH = actual` (no quotes)
B6	Why seasonal naive came last: network growth in a year	`WEEKS_BACK = 52`
C1	Tickets per 1,000 terminals, by month	`DIVIDE_BY = train_terminals` (no quotes)
C2	When does the rain arrive?	a month number, 1–12
C3	Rate seasonal naive: the rate	`series / terminals`
C4	Score it on the same 8 weeks	`HORIZON = H`
D1	The flat training average	`FLAT = train.mean()`
🌶️ **Bonus**	Which α follows the latest week closely?	`0.1`, `0.3` or `0.9`

The code you'll use

```
tickets = pd.read_csv(data_path("ops_tickets_weekly"), parse_dates=["week_start"],
                      index_col="week_start").asfreq("W-MON")    # one row per Monday
y = tickets["tickets_raised"]
train, actual = y.iloc[:-H], y.iloc[-H:]                         # learn from the past,
judge on the last H
assert train.index.max() < actual.index.min()                    # no overlap, ever

naive(train, H)                                  # every baseline: (training series,
horizon) -> H numbers
rate_seasonal_naive(train, train_terminals, H)   # the rate version also needs the
terminals
mae(forecast, actual)                            # the typical miss, in tickets
```

Blind means every forecast is made on the last training week (27 Oct 2025). The forecast for 22 December can't use the real number from 15 December.

When things go wrong: error → fix

You see	It means	Fix
`AssertionError: ✍️ Fill in … first`	A `___` is still there	Replace it, run the cell again
`AssertionError: ✍️ Finish … first` / `Get B1 to B4 to ✅ first`	An earlier task isn't done or isn't ✅ yet	Go back, fix it, then come back
`❌ Not yet … (Python said: TypeError)` under a function	The `___` inside the function is still there, or the wrong type	Fill the blank in the `def` line or the body, run the cell again
`NameError: name 'W' is not defined`	You typed `W-MON` without quotes	Frequencies are text: `"W-MON"`
A1 shows 103 weeks and 206 gaps	You used `"W"` (weeks ending on **Sunday**)	Every date is a Monday: `"W-MON"`
`NameError: name 'mae' is not defined`	You skipped the ▶️ `def mae` cell	Run it (top of Part B)
`AssertionError: ✍️ Use the variable itself, without quotes`	You typed `"train_terminals"`	Type `train_terminals` (a variable, not text)
C4 shows 52 numbers instead of 8 / a length error	`HORIZON` isn't 8	`HORIZON = H`
Everything is weird	–	**Runtime → Restart session and run all**

Rules of thumb you'll keep forever

1. **Baselines first.** Before any model, forecast with the cheap ones. A model that can't beat them has not earned its place.
2. **Lock the test weeks away** and forecast them blind, from one origin. No peeking, no expo.
3. **Same weeks, same scorer** (MAE) for every forecast you compare.
4. **Say MAE in words:** "off by about N tickets on a typical week".
5. **Divide by the size you already know** (terminals, drive, rooms). A rate repeats; a count grows.
6. **Beat the BEST baseline**, not the one with the fanciest name.
7. **Don't choose settings (like α) by looking at the test weeks.** That's expo.
8. **One split is one result.** Test on many splits before you trust it (Week 6 · Day 1).

🏁 **Finished?** Run `scoreboard()`, then the 🌶️ α task. For your project: build this same table (your baselines, same test period, same MAE) for **Checkpoint 4** next session.